

ML EVALUATION

Gamma/Hadron Classification

GROUP 32

MOTIVATION



REAL-WORLD APPLICATION OF ML



SUPPORT STUDY OF HIGH-ENERGY PHENOMENA SUCH AS SUPERNOVA OR BLACK HOLES



HELPS ASTRONOMERS DISTINGUISH SIGNALS FROM NOISE



OPPORTUNITY TO COMPARE CLASSIFICATION ALGORITHMS ON SCIENTIFIC DATA



PERSONAL INTEREST IN THE FIELD OF AI AND ASTRONOMY



DEMONSTRATING HOW ML CAN CONTRIBUTE TO SCIENTIFIC RESEARCH

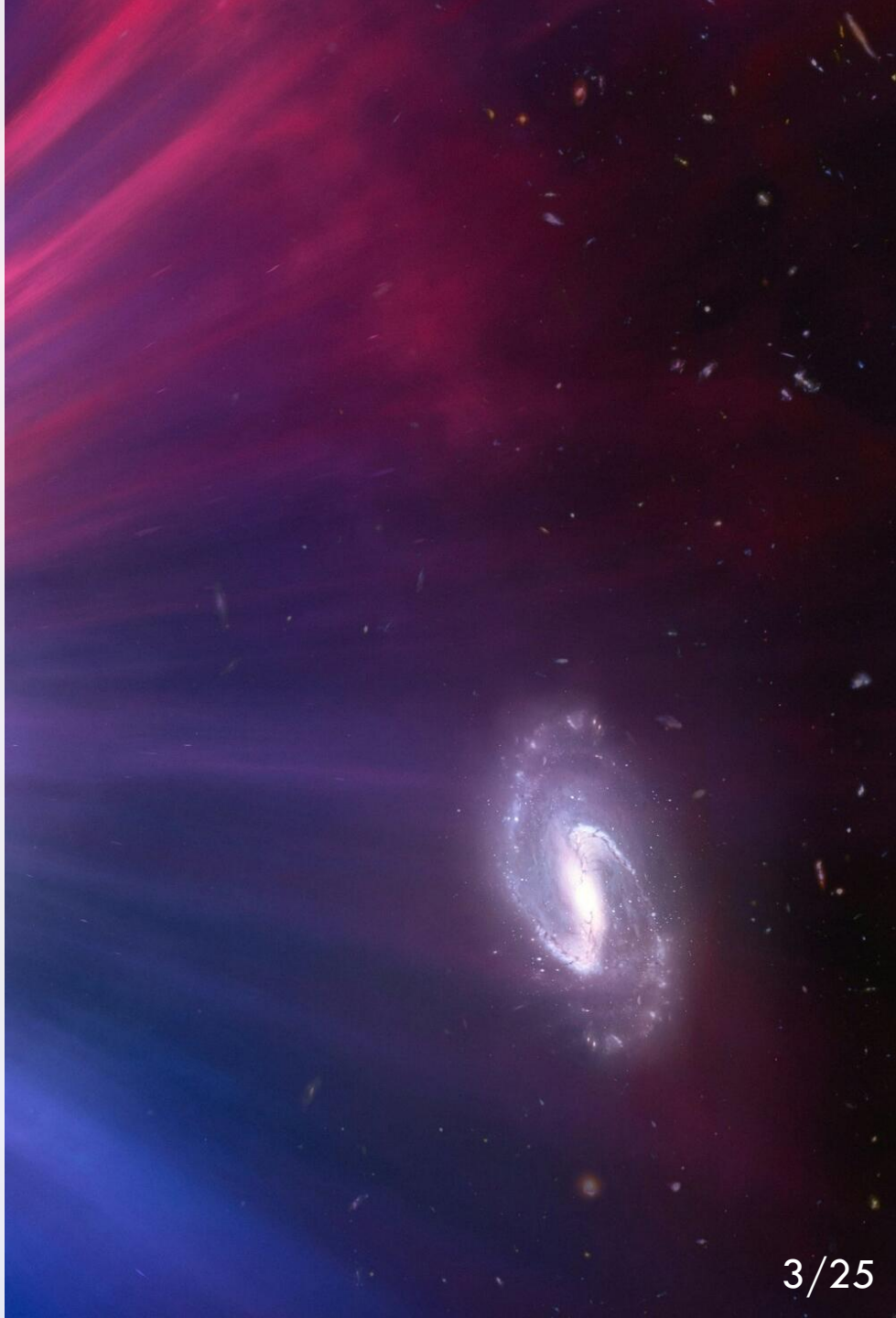
WHY SEPARATE GAMMA RAYS FROM HADRONS?

The MAGIC Cherenkov telescope records faint light of air showers in the atmosphere

Real gamma-ray photons carry astrophysical signals that humanity wants to study

Hadronic cosmic-ray showers look similar in the raw data but are background noise – far more numerous in real life

Which classifier recovers gamma events best?



METHOD

MODELS

kNN (baseline)

SVM-RBF (non-linear boundary)

Random Forest (imbalance-robust +feature importance)

Split & Leakage

- 80/20 stratified split
- Standardization & feature selection redone inside each CV fold -> no peeking at validation data

Tuning (5-fold stratified CV)

- kNN: k = 11
- SVM: C = 752.5, γ = 0.0316
- RF: 300 trees, depth 20 Test set used once, at the very end

Metric

- auROC + sensitivity/specificity, not accuracy (imbalance -> misleading)

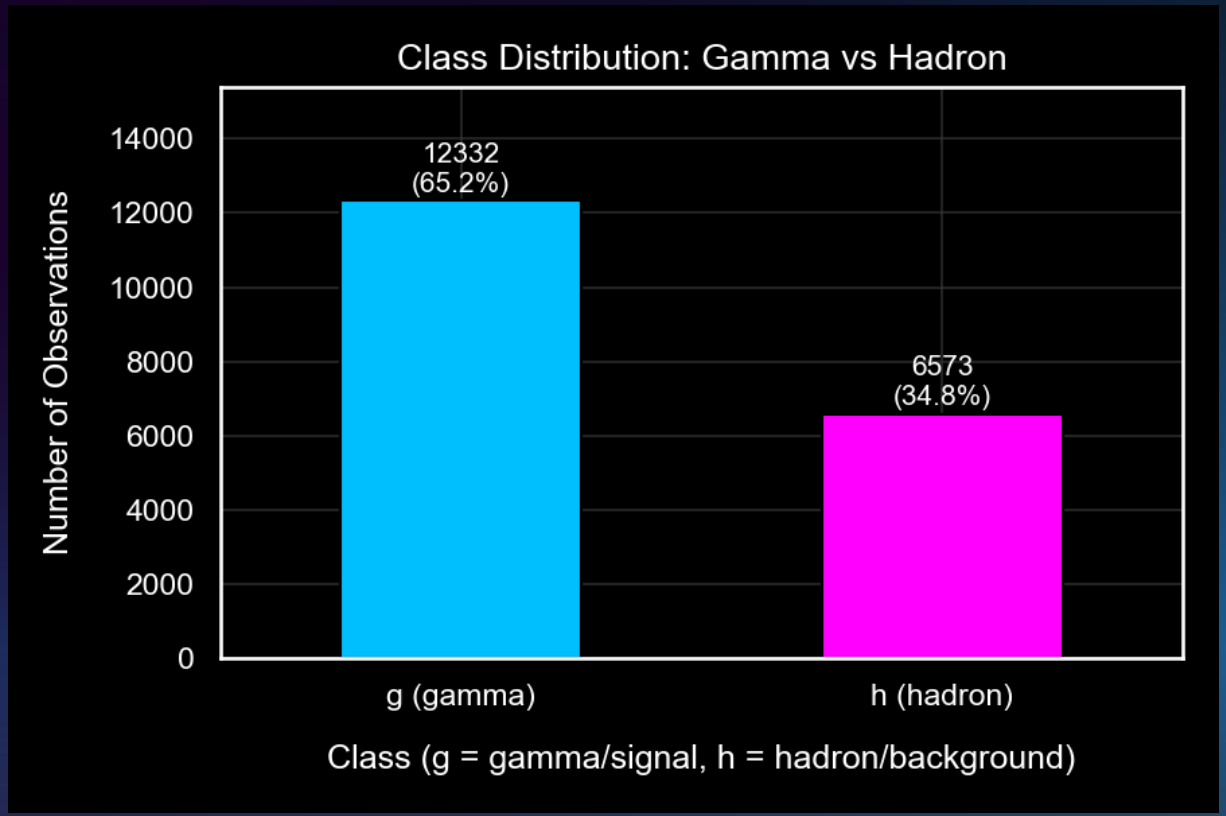
EXPLORATORY DATA ANALYSIS

Class
g = gamma
h = hadron

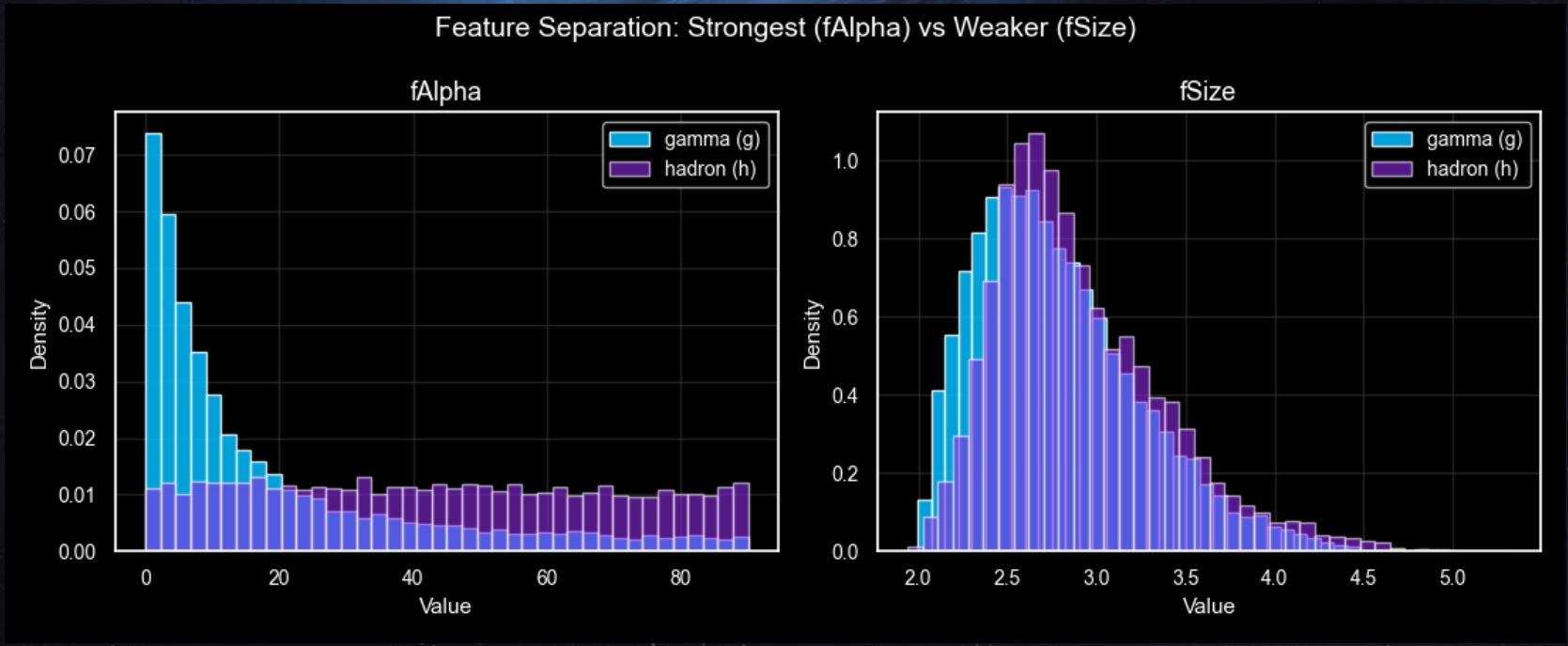
- fLength
- fWidth
- fSize
- fConc
- fConc1
- fAsym
- fM3Long
- fM3Trans
- fAlpha
- fDist

18,905
Samples

10
features



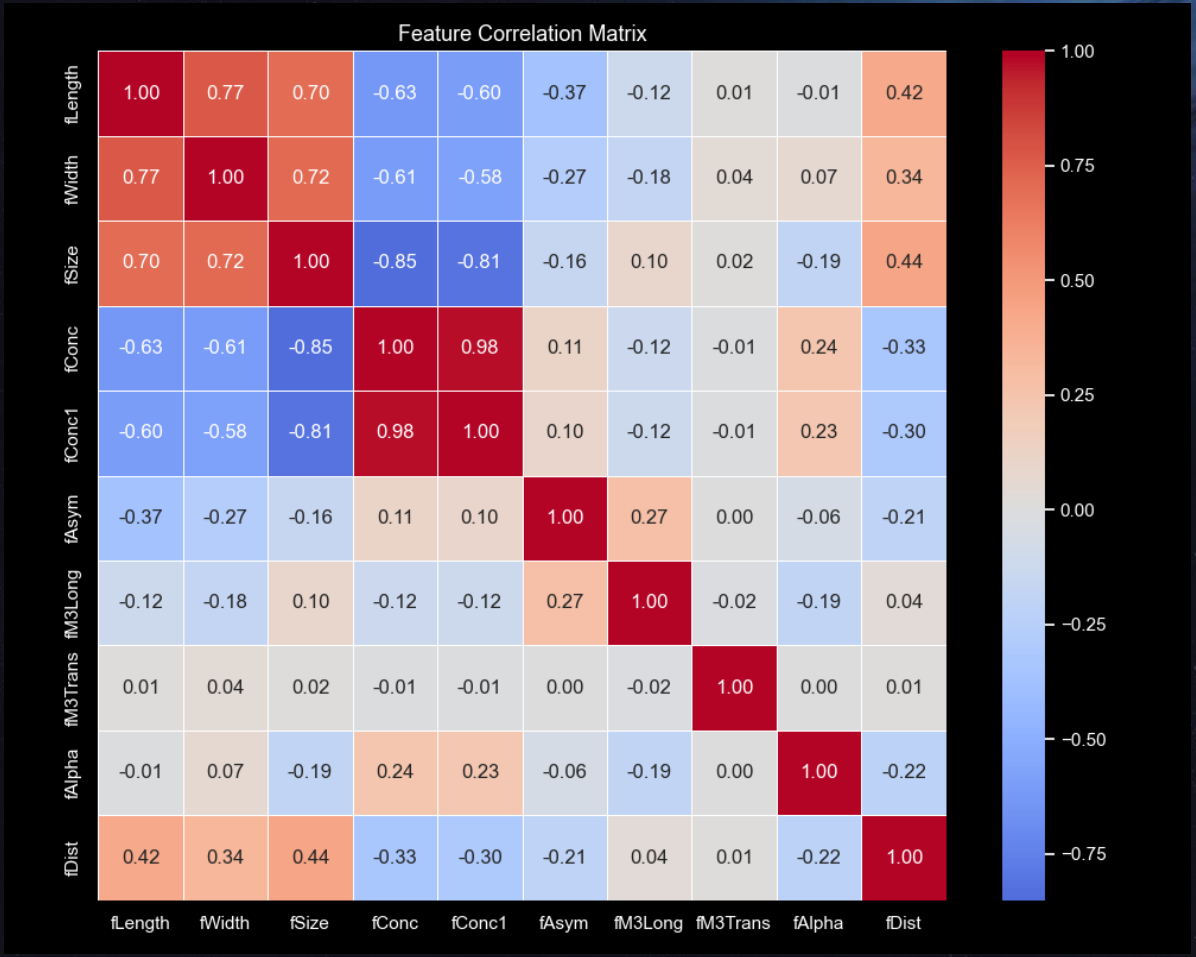
WHICH FEATURES SEPARATE THE CLASSES?



fAlpha is the strongest discriminator – gamma showers point back to the source, hadron showers don‘t

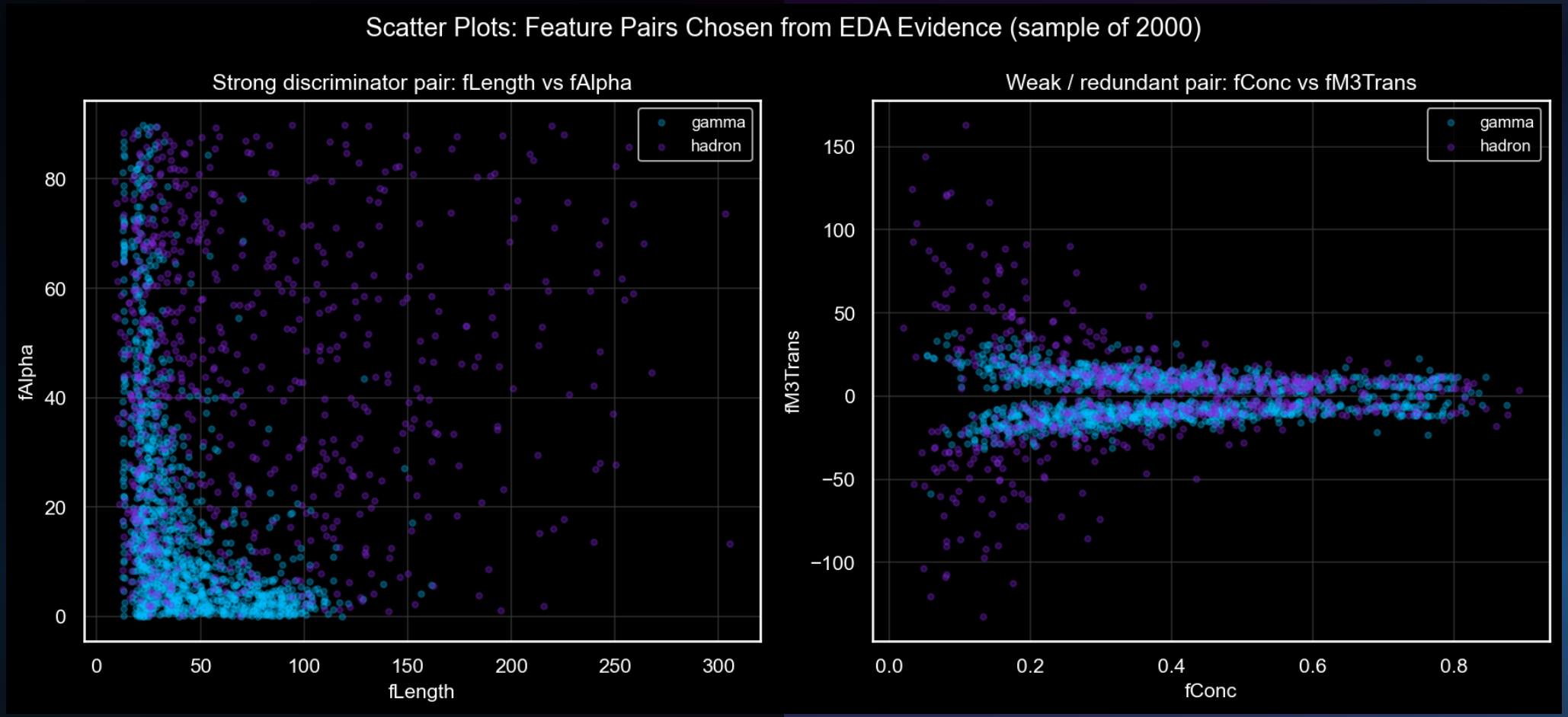
fSize separates moderately, despite the shower brightness

FEATURE REDUNDANCY CHECK



- fConc and fConc1 have strong correlation – expected, both describe light concentration
- fConc and fSize have strong correlation
- Correlation with another feature is not the same as usefulness for classification – this motivates letting each model select it‘s own features rather than assuming

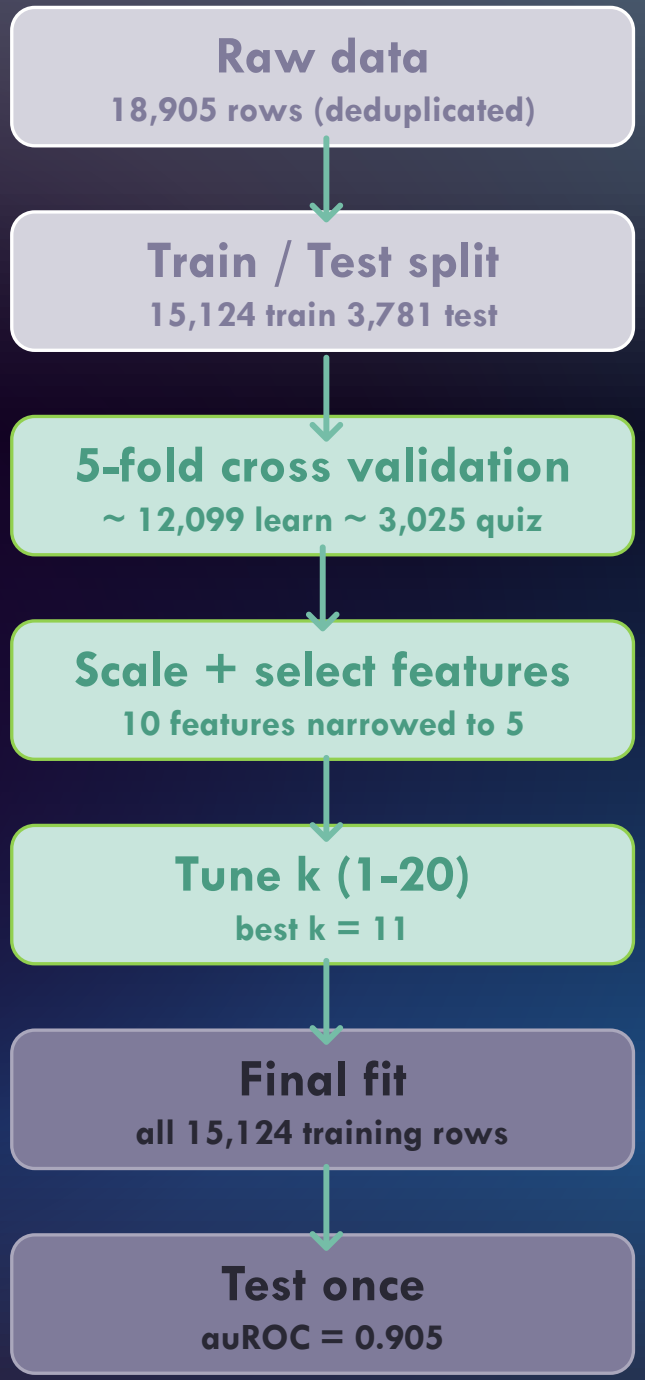
STRONG VS. WEAK FEATURE PAIRS



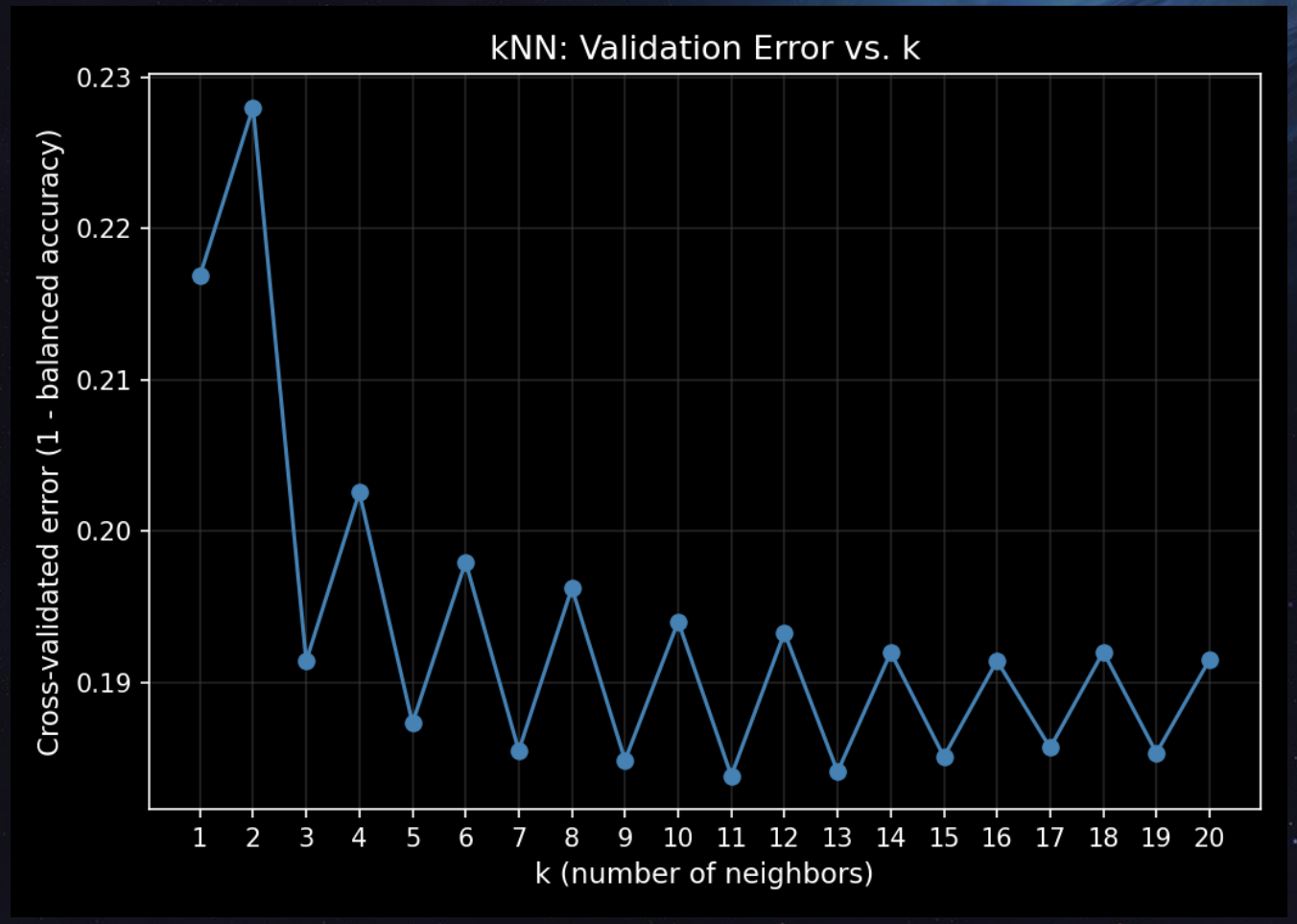
KNN PIPELINE

Selected features (5)
fLength
fWidth
fSize
fConc
fAlpha

Dropped features (5)
fConc1
fAsym
fM3Long
fM3Trans
fDist



KNN: FEATURE SELECTION & K TUNING



Backward elimination (wrapper method) kept only 5 of 10 features:

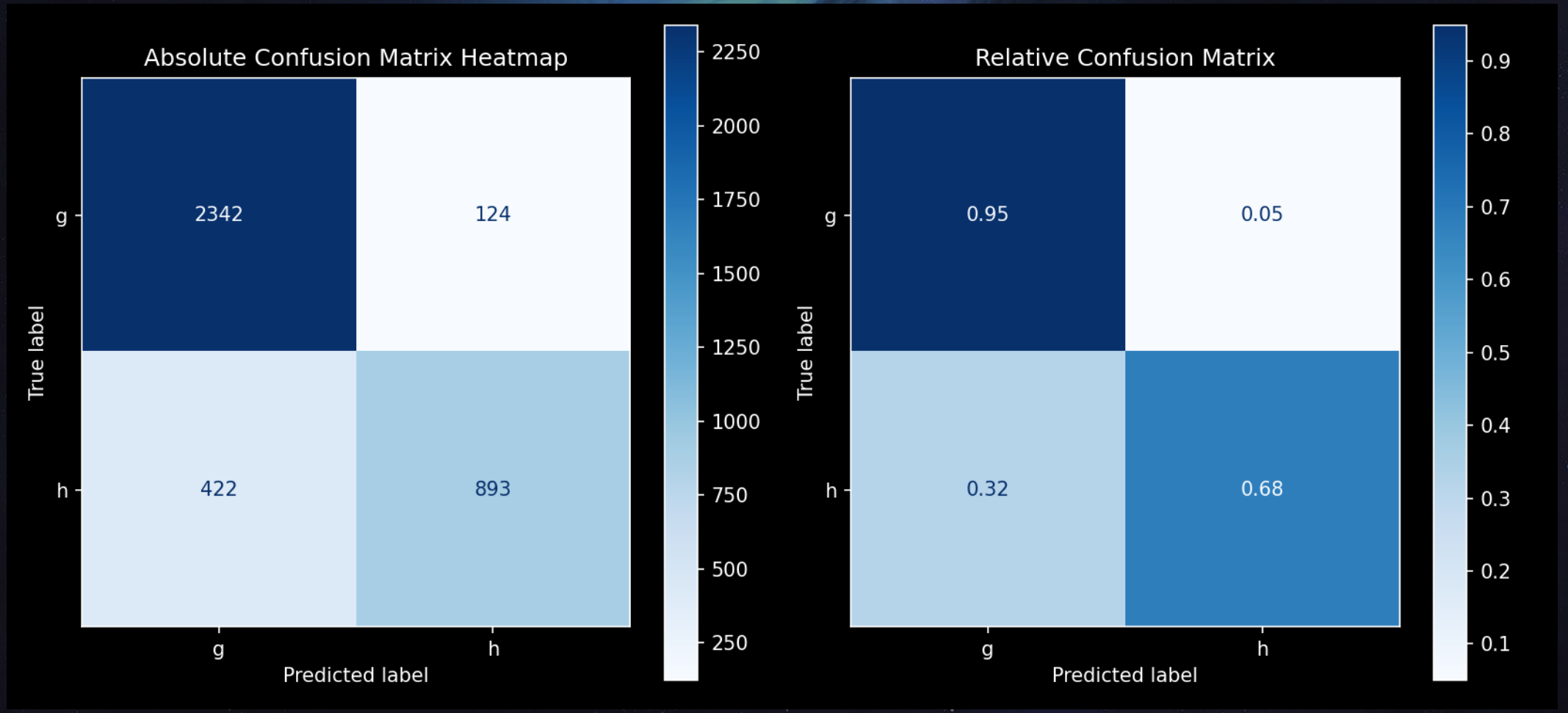
fLength, fWidth, fSize, fConc, fAlpha

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kNN relies on distances between points, so irrelevant features can hurt its performance more than other models

Best k = 11, chosen via nested cross-validation

KNN: RESULTS


Absolute Confusion Matrix Heatmap
2342
124
422
893
g
h
True label
Predicted label
2250
2000
1750
1500
1250
1000
750
500
250
Relative Confusion Matrix
0.95
0.05
0.32
0.68
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1

KNN: RESULTS

Sensitivity (gamma)

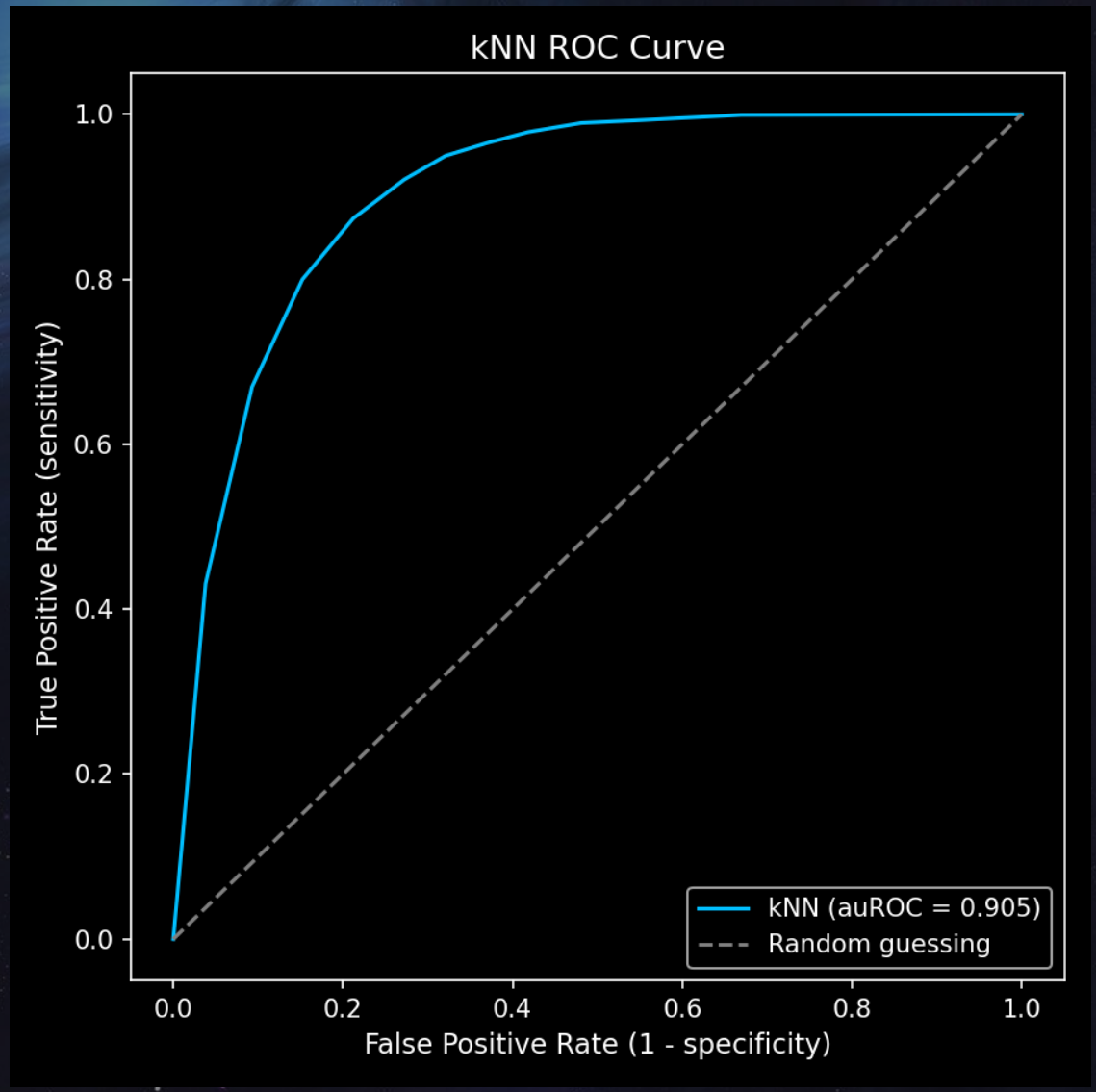
95.0%

Specificity (hadron)

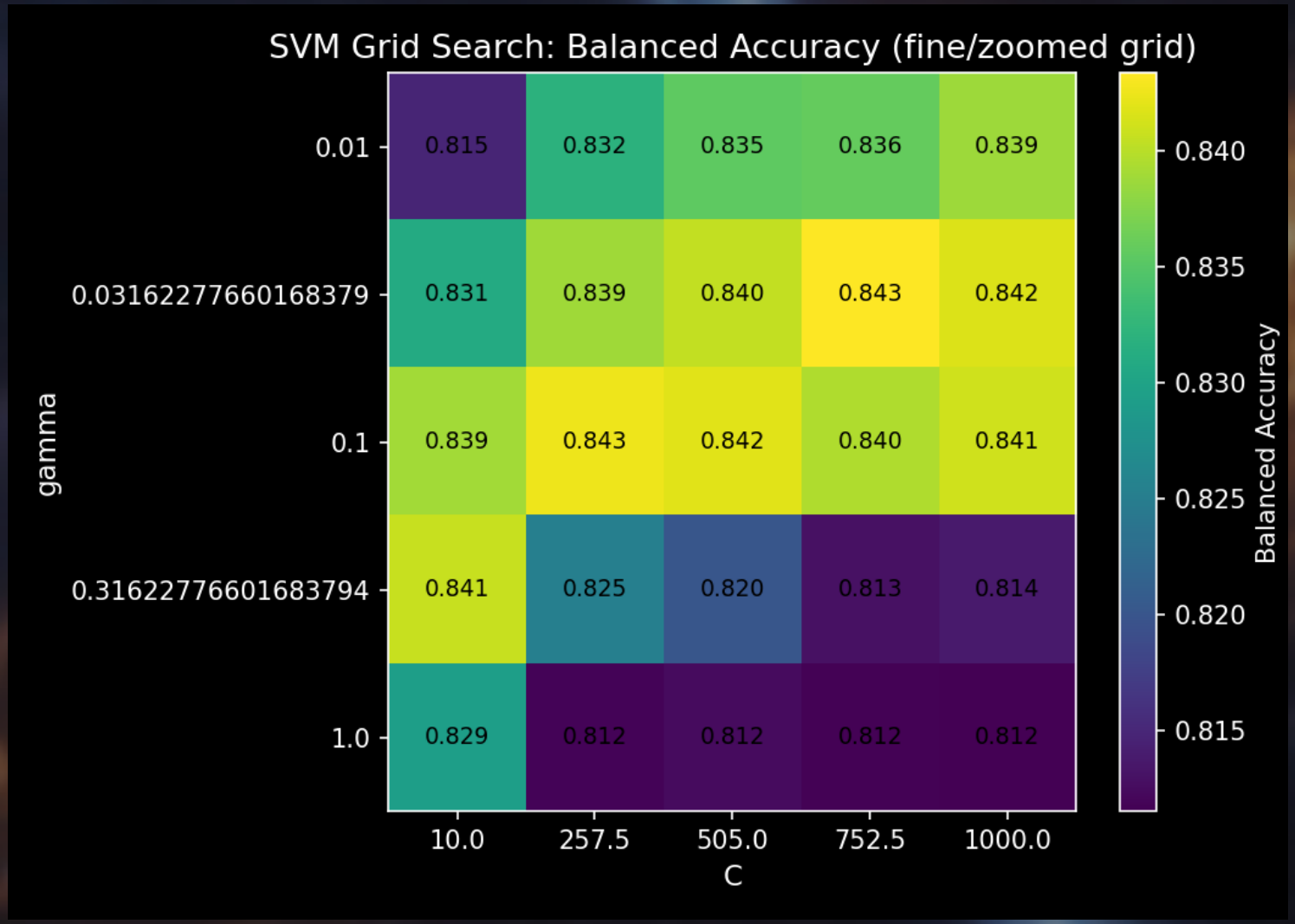
67.9%

auROC

0.905



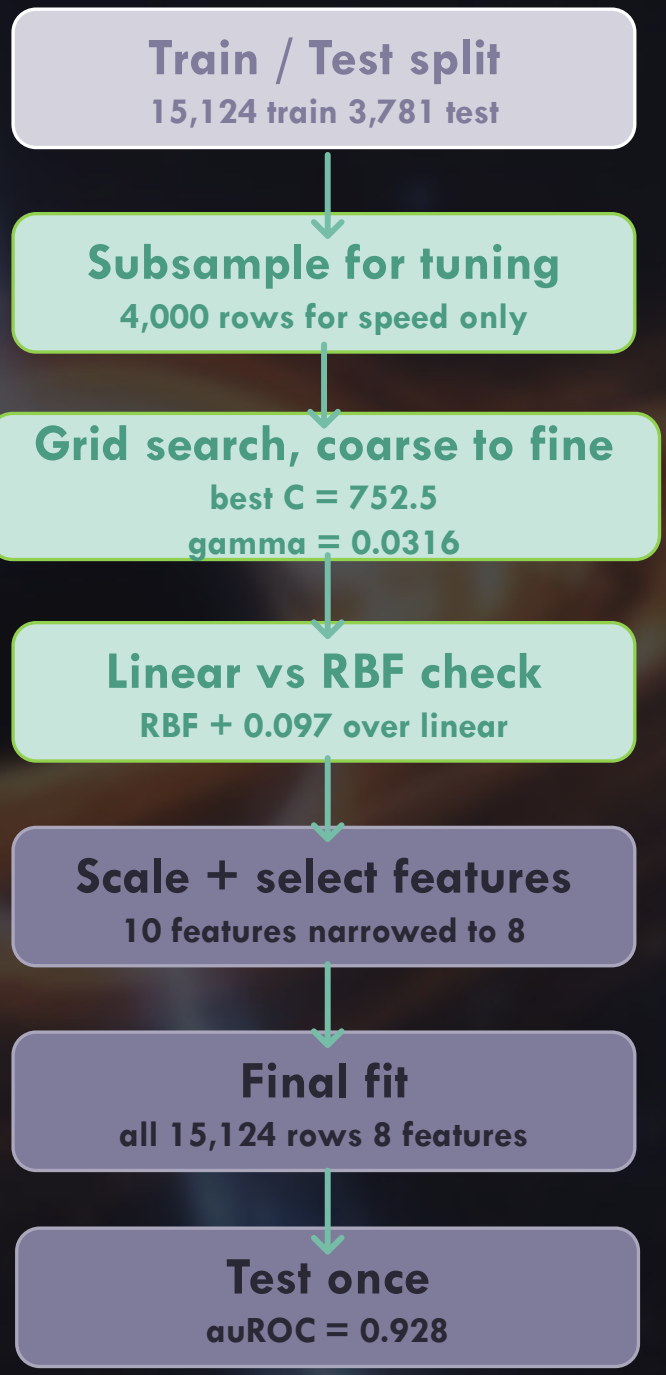
SVM: KERNEL CHOICE & TUNING



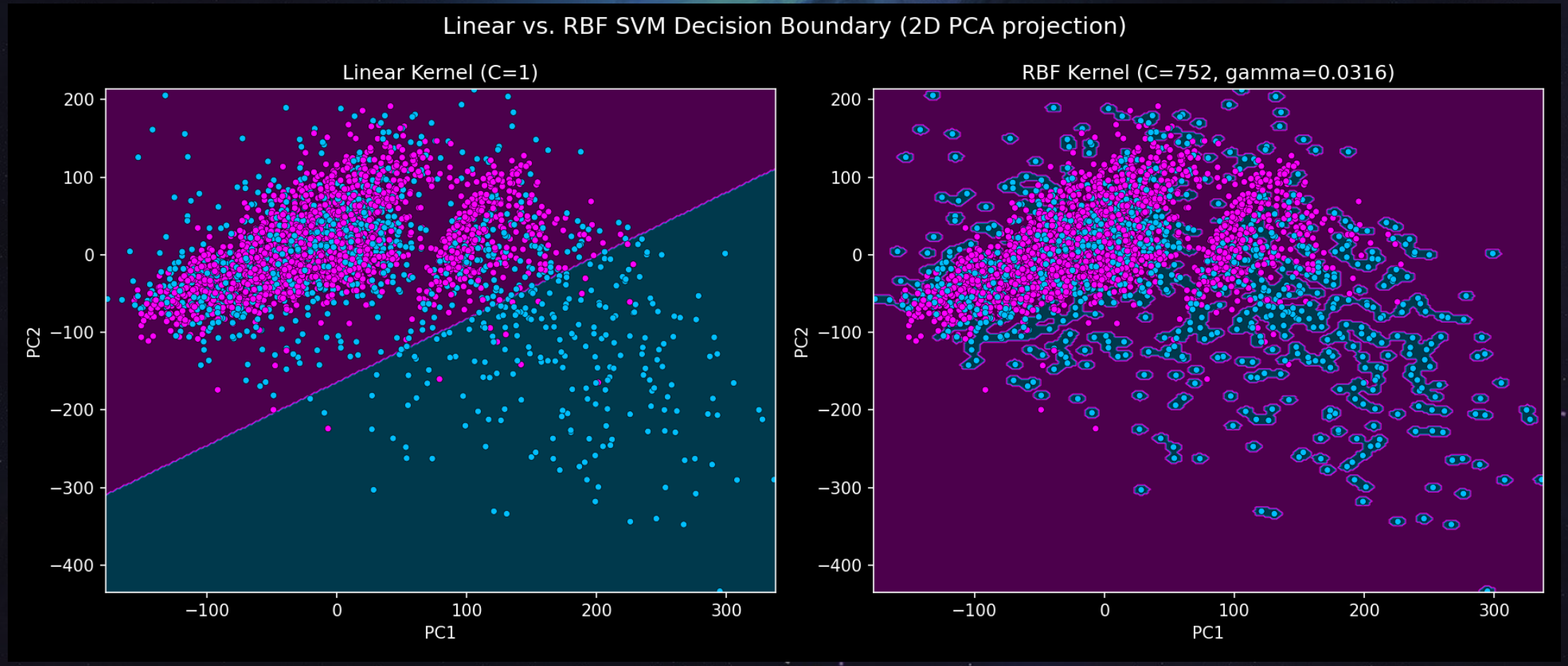
SVM PIPELINE & FEAT. SELECTION

SVM Backward Elimination (10 ⟶ 8 features)

fLength	✓
fWidth	✓
fSize	✓
fConc	✓
~~fConc1~~	✗
fAsym	✓
fM3Long	✓
~~fM3Trans~~	✗
fAlpha	✓
fDist	✓

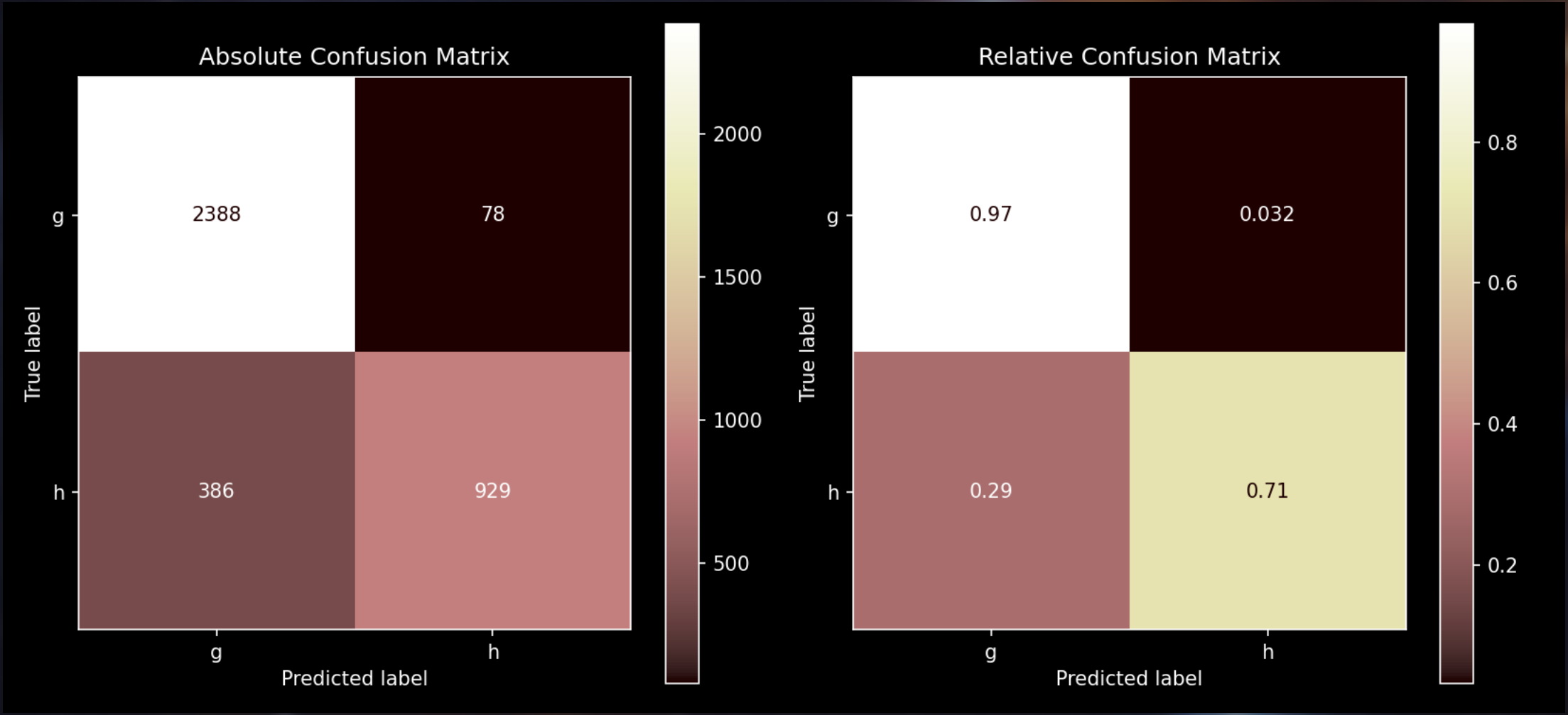


SVM: LINEAR VS RBF KERNEL COMPARISON



PCA compresses the 10 features into 2 dimensions by the variance of the observations, while keeping as much information as possible so we can visualize the data

SVM: RESULTS


Absolute Confusion Matrix
g
h
True label
2388
78
386
929
g
h
Predicted label
2000
1500
1000
500
Relative Confusion Matrix
g
h
True label
0.97
0.032
0.29
0.71
g
h
Predicted label
0.8
0.6
0.4
0.2

SVM: RESULTS

Sensitivity (gamma)

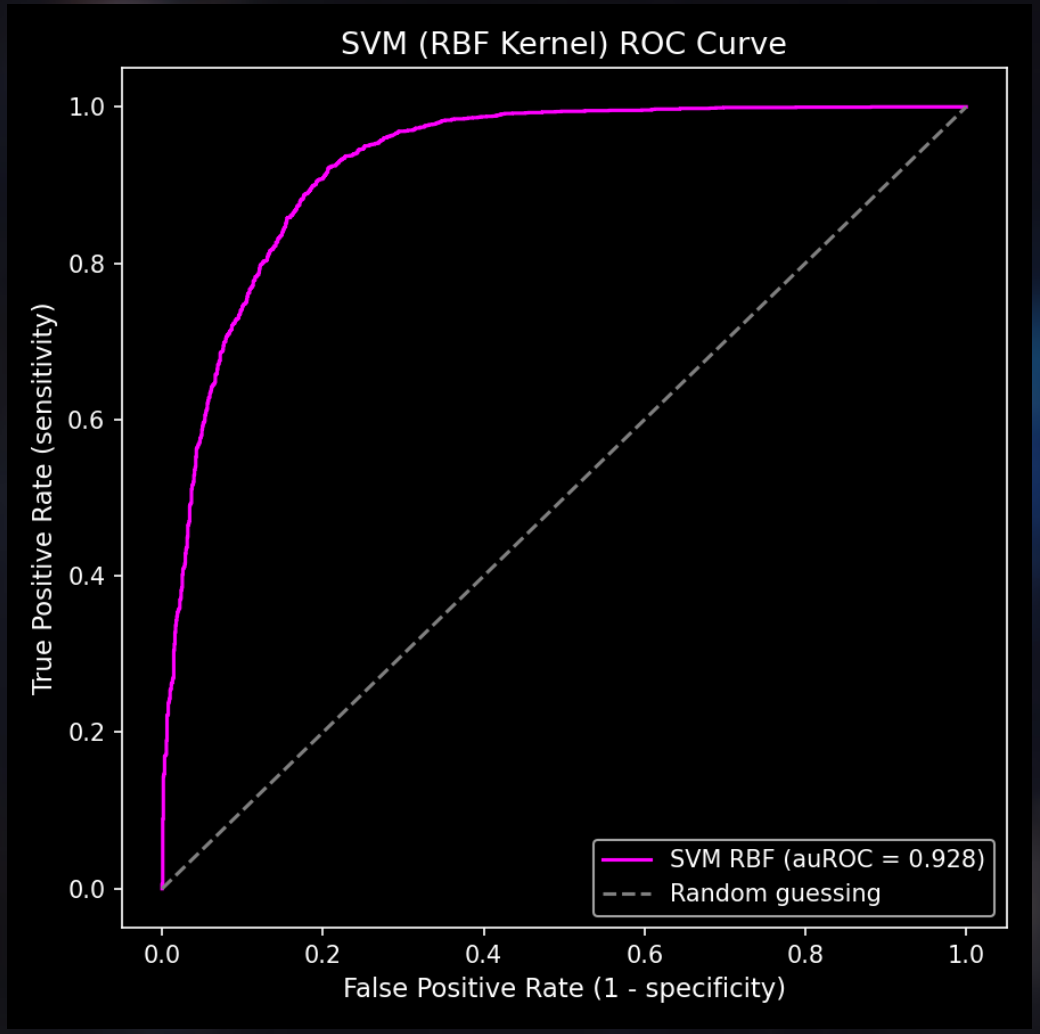
96.8%

Specificity (hadron)

70.6%

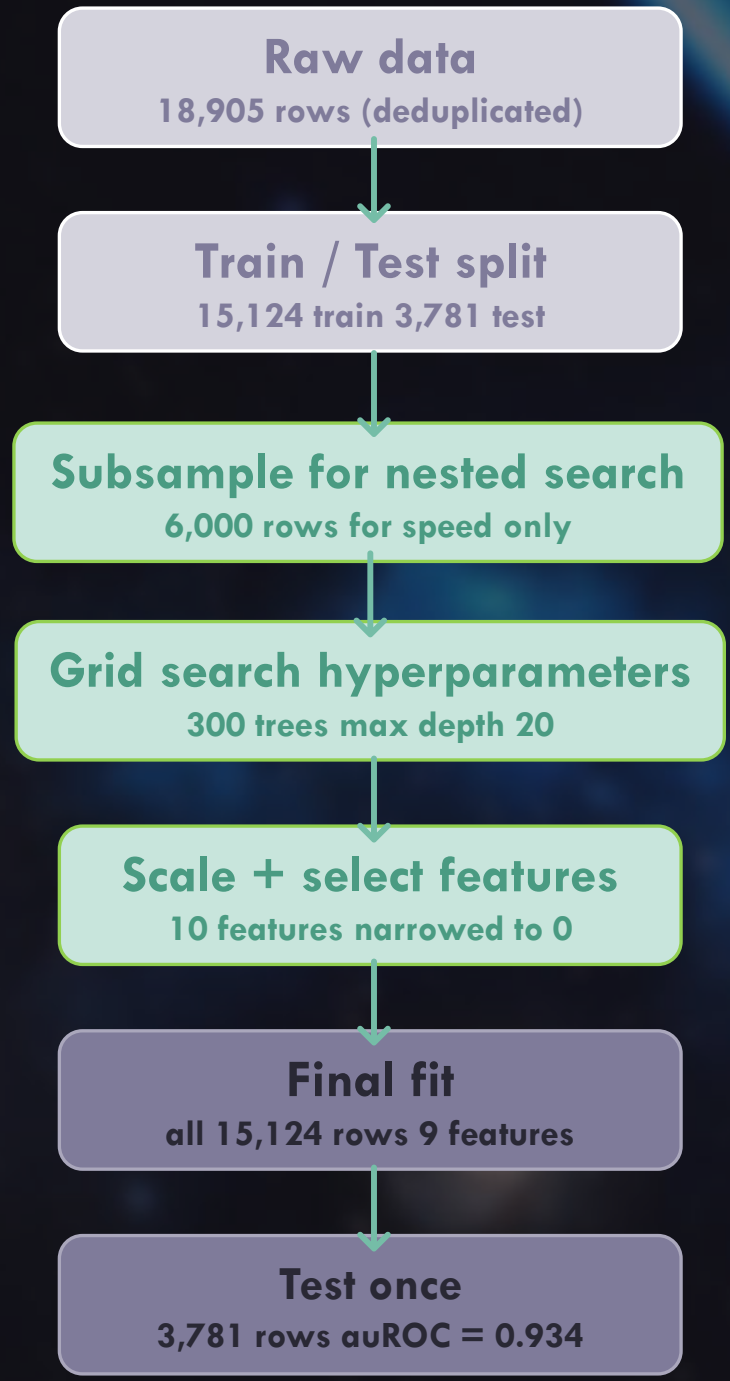
auROC

0.928

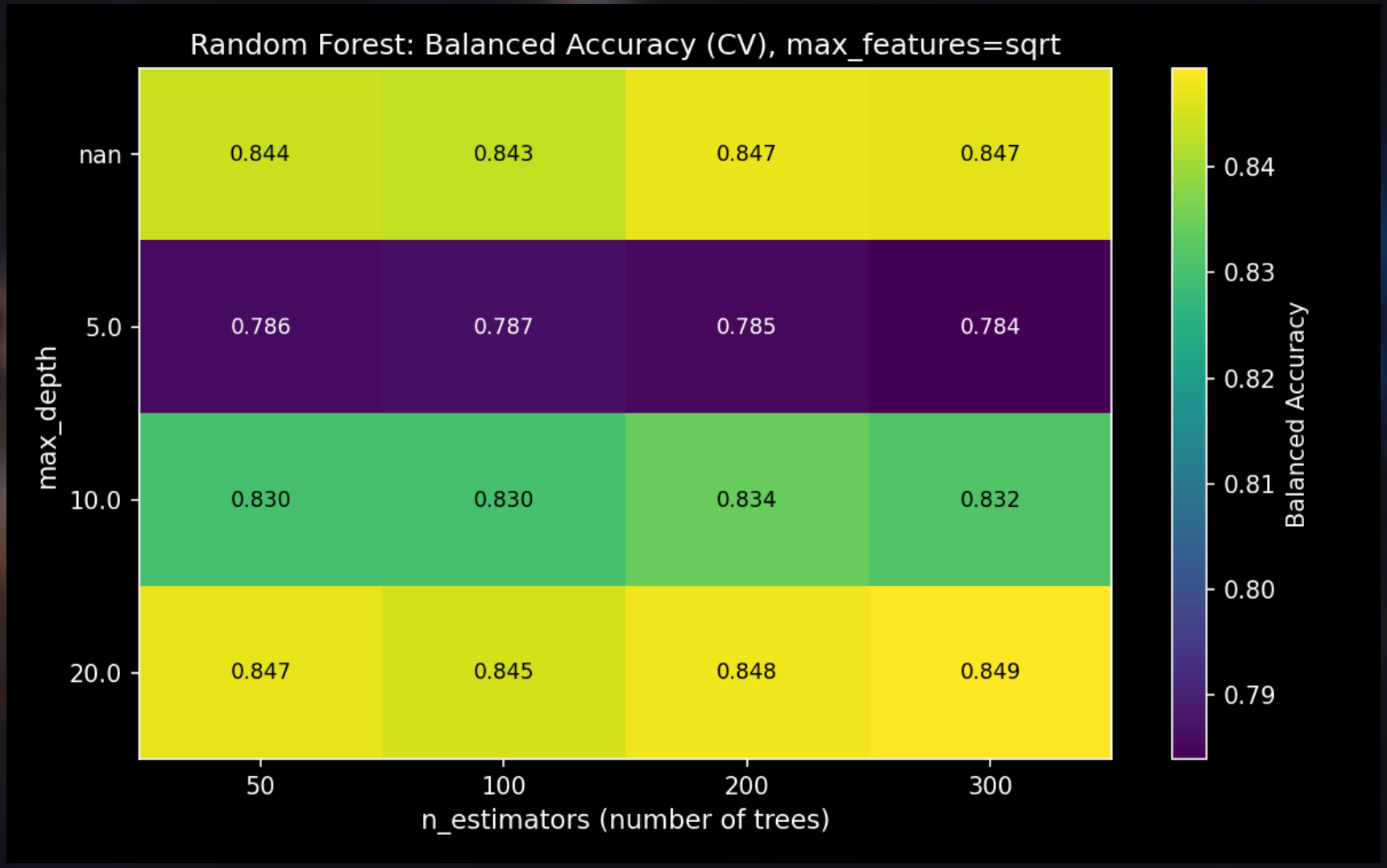


RANDOM FOREST PIPELINE + B TREES & MAX TREE DEPTH

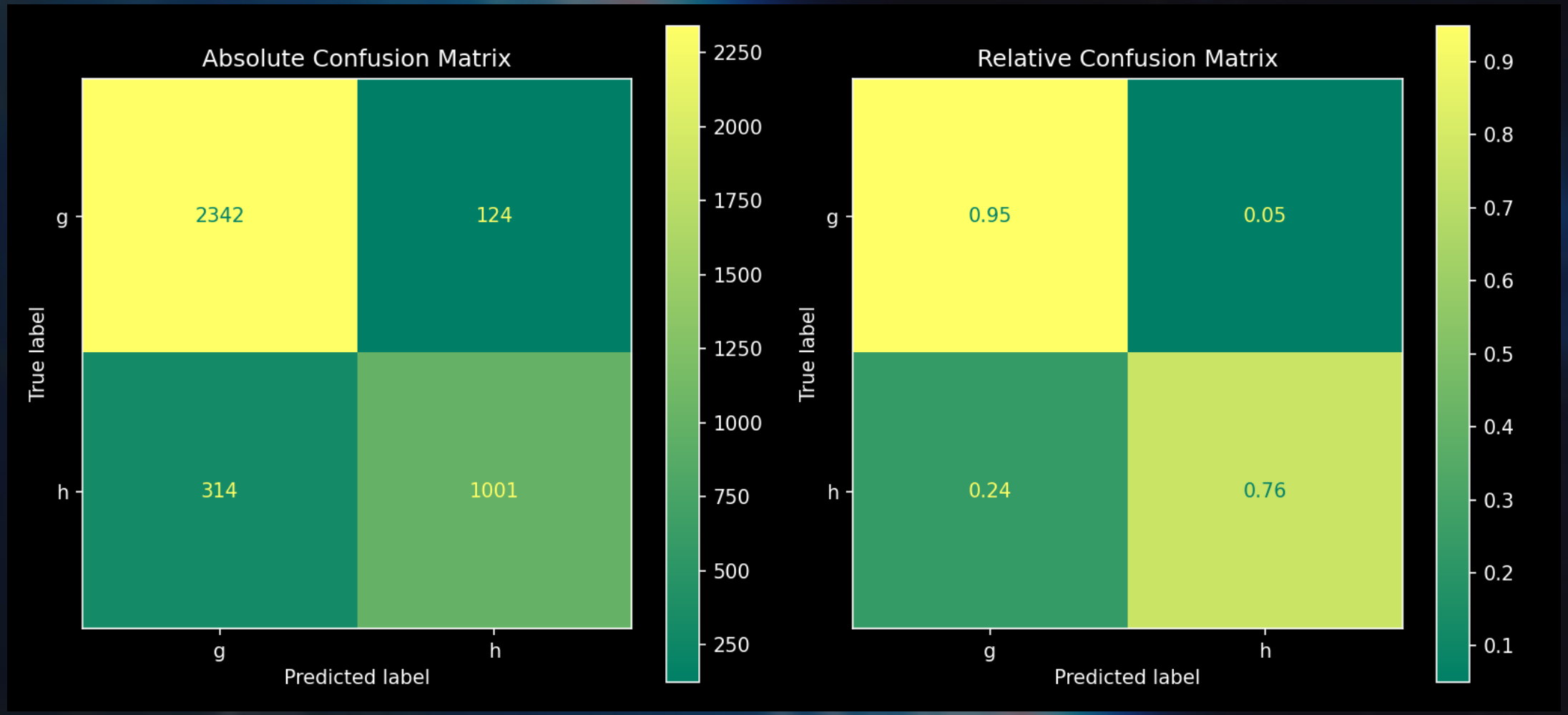
fLength	✓
fWidth	✓
fSize	✓
fConc	✓
fConc1	✓
fAsym	✓
fM3Long	✓
~~fM3Trans~~	✗
fAlpha	✓
fDist	✓



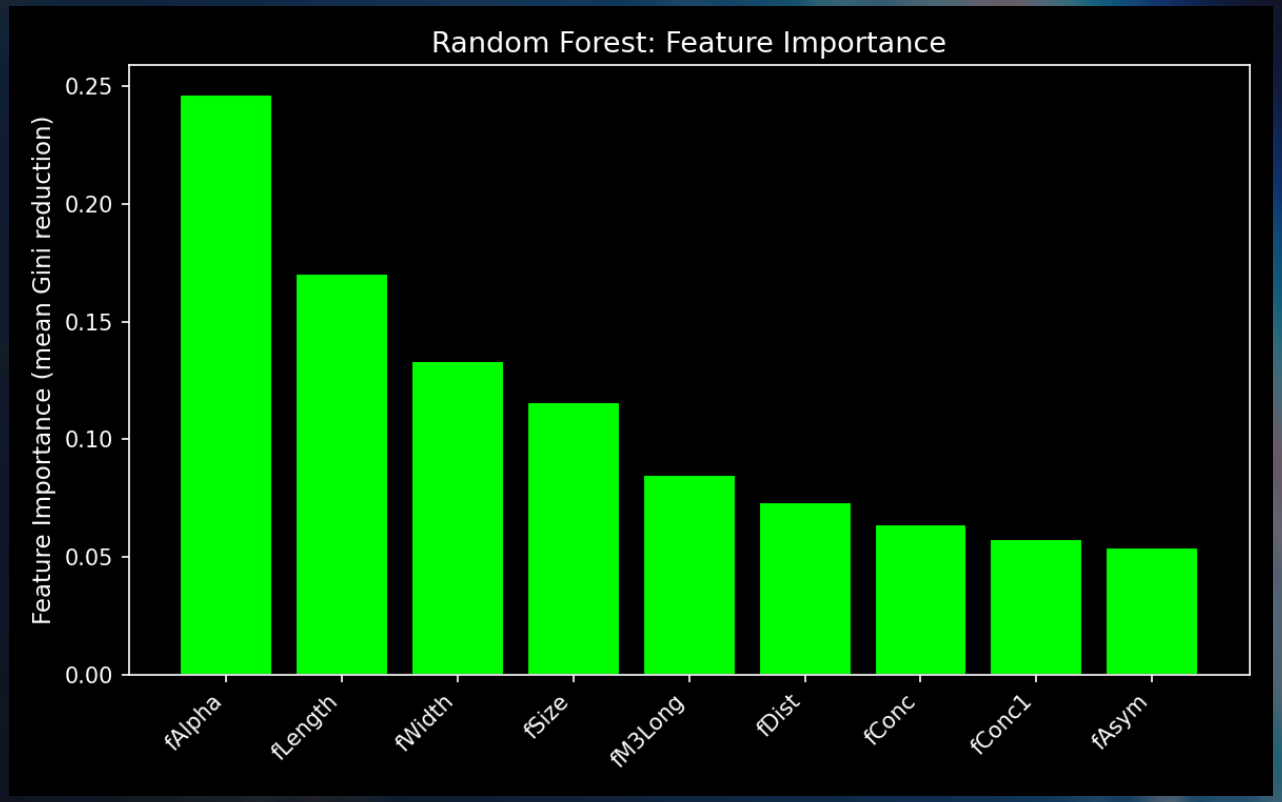
RANDOM FOREST PIPELINE + B TREES & MAX TREE DEPTH



RANDOM FOREST: CONFUSION MATRIX



RANDOM FOREST: RESULTS – FEATURE IMPORTANCE



Random forest calculates feature importance by adding each feature‘s impurity reduction across all trees and dividing it by the total impurity reduction, making all importance scores add up to 1

RANDOM FOREST: ROC & AUROC

Sensitivity (gamma)

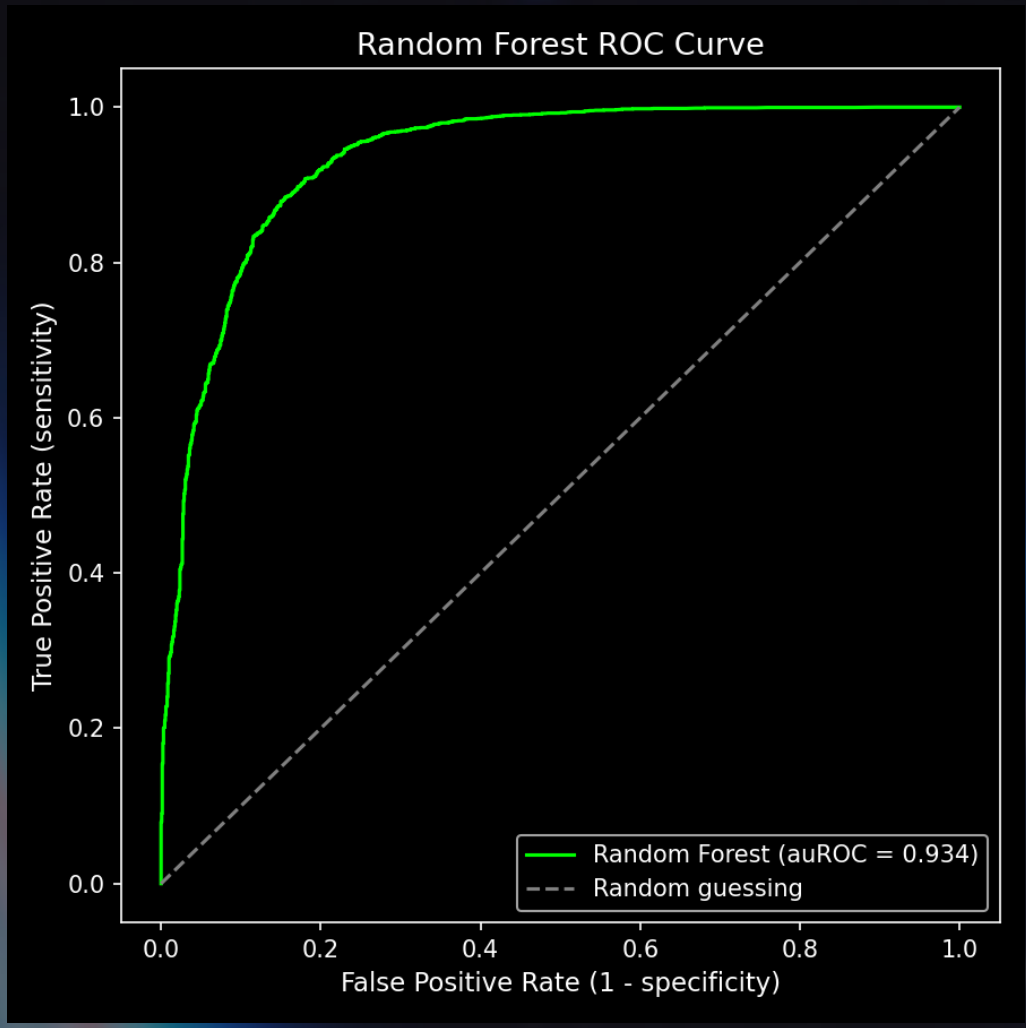
95.0%

Specificity (hadron)

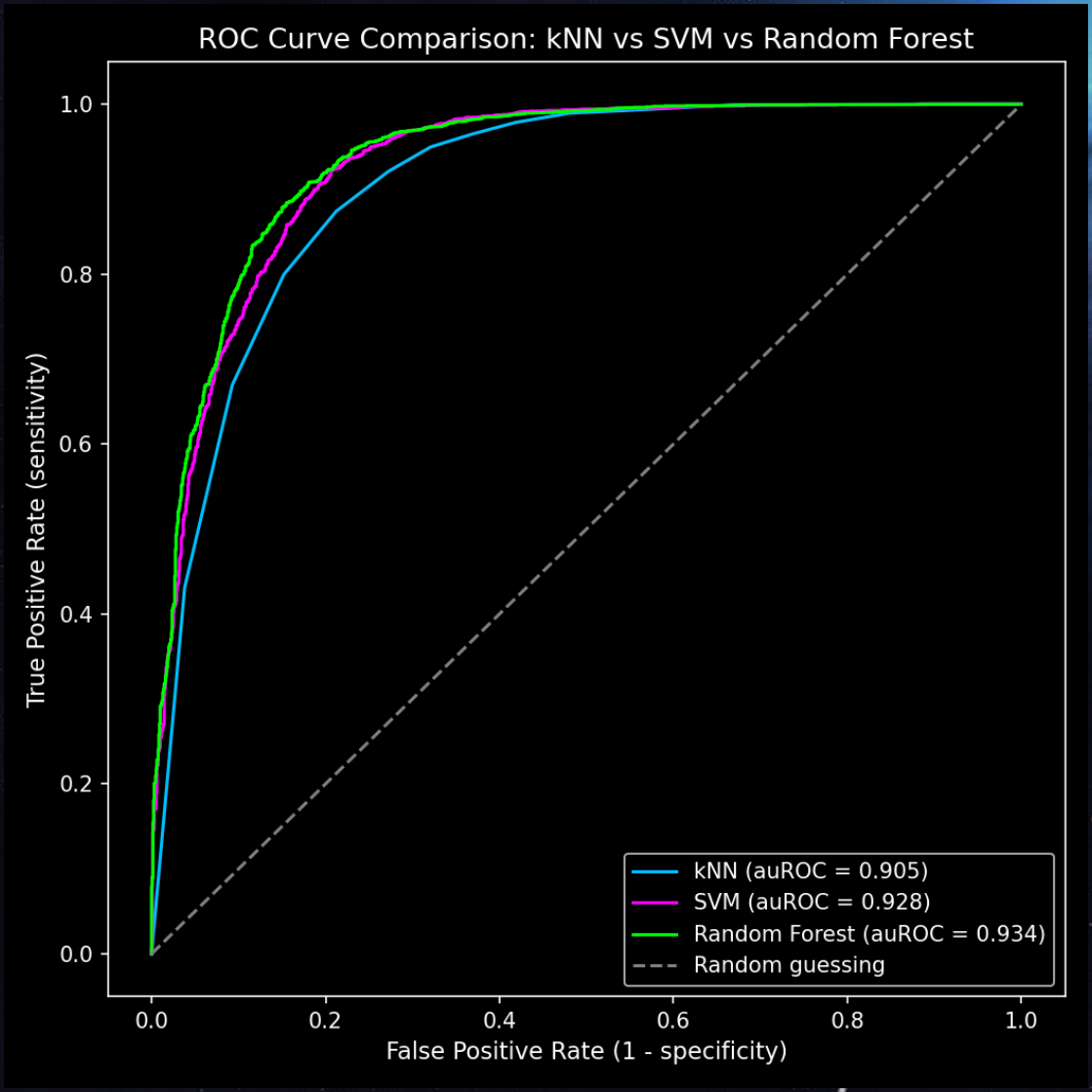
76.1%

auROC

0.934



COMPARING ALL 3



Model	auROC	Sensitivity	Specificity	Misclassified (of 3,781)
kNN	0.905	0.950	0.679	546
SVM	0.928	0.968	0.706	464
RF	0.934	0.950	0.761	438

RESULTS & DISCUSSION

Random Forest wins with the highest auROC score, showing that it separates gamma and hadron events most efficiently

Class imbalance (~65/35) makes it so accuracy is not the right metric, meaning that sensitivity, specificity and auROC were used instead

Feature importance shows that fAlpha is the most important feature, supporting our EDA hypothesis, while feature selection shows that removing fM3Trans has little effect on performance of any model

Trade-off: Random Forest is the most expensive to use; SVM is a close 2nd choice at lower computational cost

SUMMARY

Winner:

Random Forest (auROC 0.934) beats SVM (0.928) and the kNN baseline (0.905), with the best sensitivity/specificity

fAlpha confirmed:

our EDA hypothesis was backed by all 3 model's feature selection and RF's feature importance (fAlpha leads at 0.247, ahead of fLength at 0.170)

Limitation:

all 3 models failed´on the same 308 test events, most of which are hadrons. Because almost all of them don't have extreme feature values, these are likely normal but overlapping cases that are difficult for any of the models to classify correctly

Outlook:

with more time, target those 308 shared failures directly, instead of only optimizing overall auROC


THANK YOU
GROUP 32